

Diagrammatic (∞, n) -categories

Clémence Chanavat

j.w.w Amar Hadzihanovic

Tallinn University of Technology

Atom and molecules

A **molecule** is a combinatorial datum encoding the shape of a higher categorical pasting diagram. A molecule U :

- is an **oriented** and graded poset;
- has for each $\alpha \in \{-, +\}$ and $n \in \mathbb{N}$, a **input/output n -boundary**, $\partial_n^\alpha U$, which is a closed subset of U .

Molecule can be **pasted** together: given U , V , and $n \in \mathbb{N}$, if $\partial_i^+ U \cong \partial_i^- V$, then the pushout

$$\begin{array}{ccc} \partial_i^+ U \cong \partial_i^- V & \hookrightarrow & V \\ \downarrow & & \downarrow \\ U & \hookrightarrow & U \#_k V \end{array}$$

is a molecule. Pasting and boundary of molecules satisfy all the axioms of strict ω -categories, and even some **higher-exchange laws** in dimension ≥ 4 . An **atom** is a molecule with a greatest cell.

Directed complexes

A **directed complex** X is a presheaf over the category of atoms. It is the combinatorial data to construct a **positive** and **regular** polygraphs:

$$\begin{array}{ccc} \coprod_{u \in \mathcal{S}_n} \partial U & \hookrightarrow & \coprod_{u \in \mathcal{S}_n} U \\ (\partial u)_{u \in \mathcal{S}_n} \downarrow & & \downarrow (u)_{u \in \mathcal{S}_n} \\ X_{n-1} & \hookrightarrow & X_n \end{array}$$

There is a construction Mol/X that produces the strict ω -category of **molecules in X** .

- If X is a graph, Mol/X is the free category on X ;
- If X is the n -dimensional simplex, then Mol/X is the n -oriental.

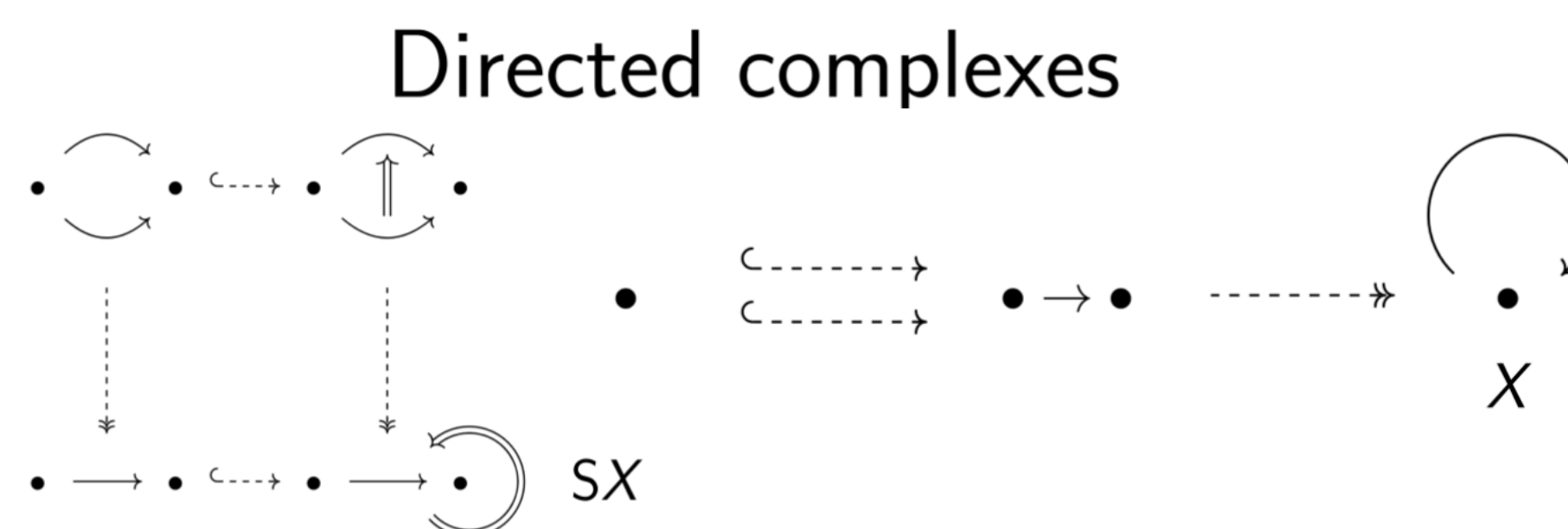
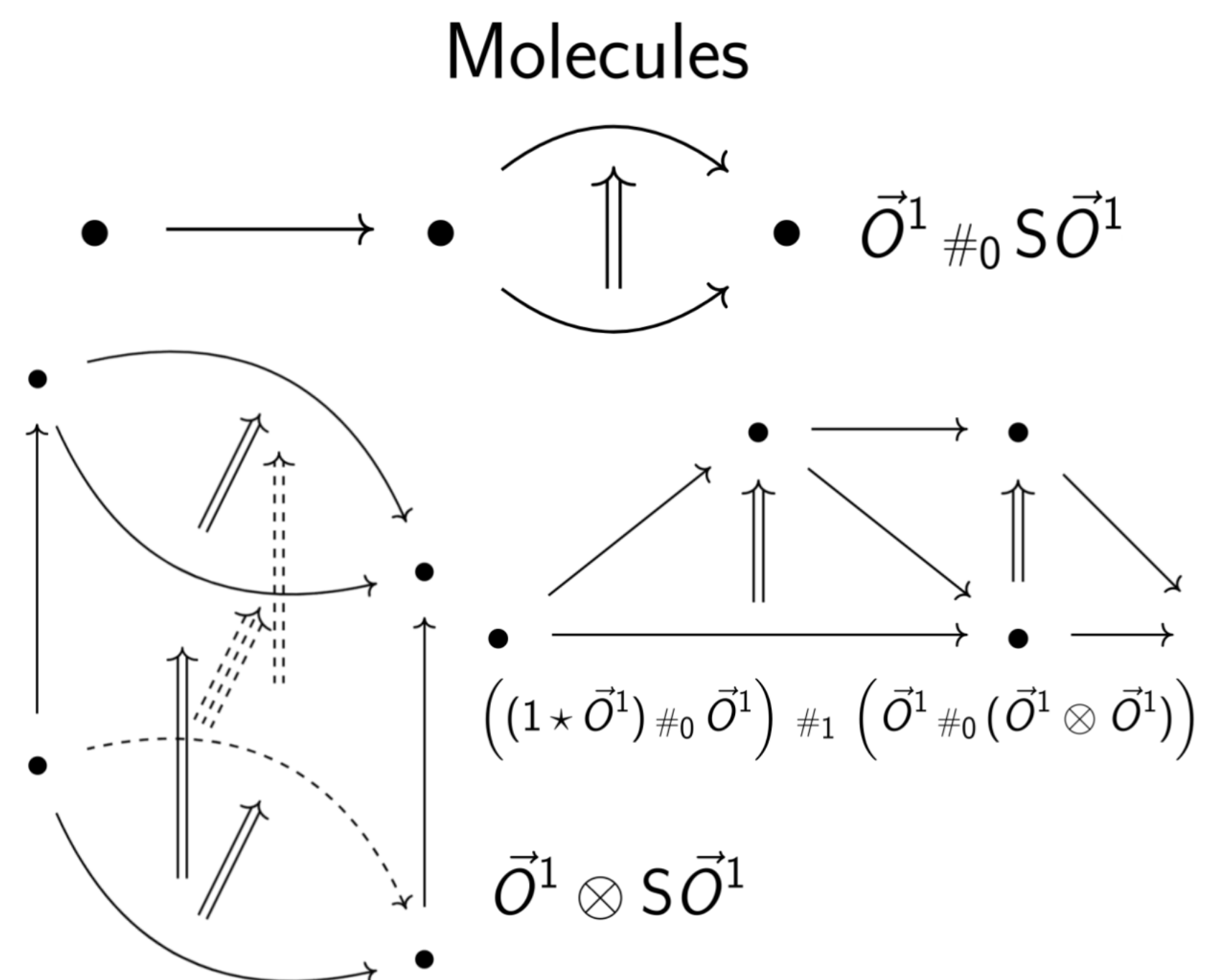
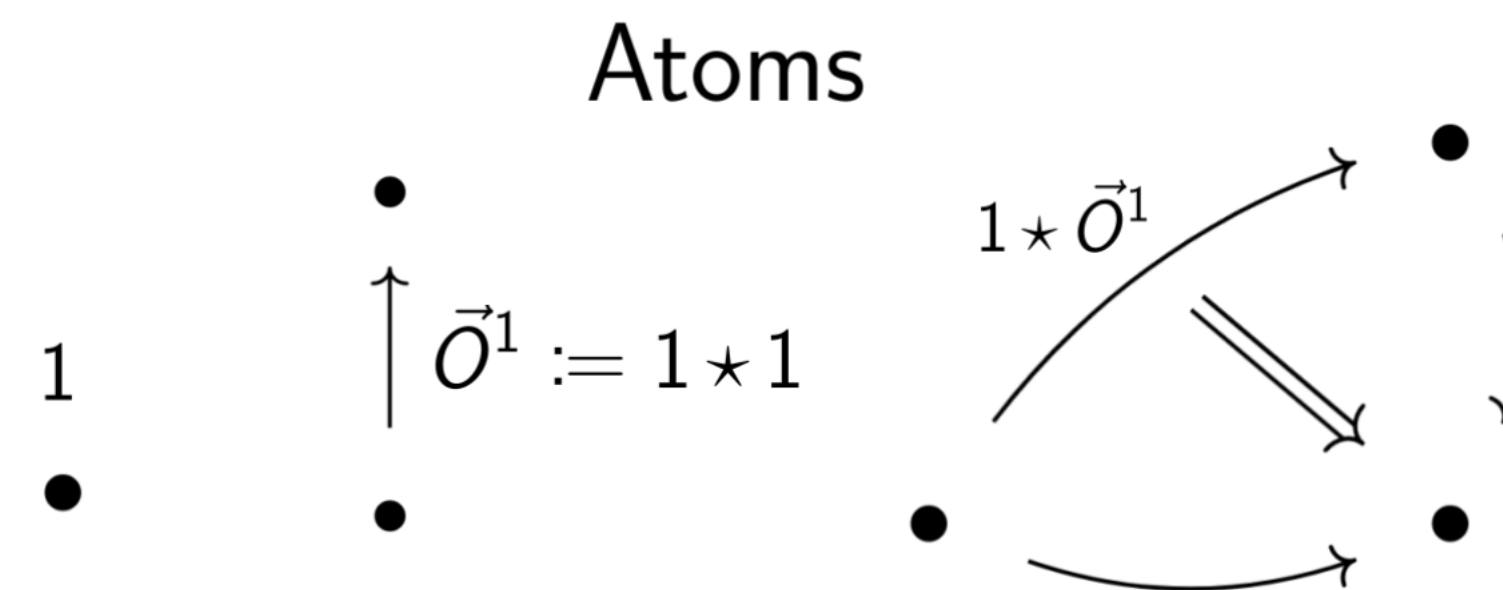
Theorem 1. under a general acyclicity condition on its molecules, Mol/X is a polygraph, and in fact, a **weak polygraph**.

Operations

Atom, molecules, and directed complexes, support natively higher categorical operations:

- Gray product $X \otimes Y$,
- join $X \star Y$,
- suspension SX ,
- duals $D_J X$, where $J \subseteq \mathbb{N}$.

Examples



In the last example:

- $Mol/X = \mathbb{BN}$
- X satisfies the acyclicity hypothesis of **Theorem 1**.

Diagrammatic (∞, n) -categories

Directed complexes may be used in several ways to construct models of $(0, n)$ and (∞, n) -categories:

1. as Segal-sheaves over the Mol/X 's and **strict functors**;
2. as presheaves over the atoms and **cellular** strict functors between them; A localisation enforces the existence of **weak composites**.

This matches the following two perspective on standard (∞, n) -categories as

- Segal-sheaves over the theta's and strict functors;
- presheaves over the (marked) orientals and cellular strict functors (i.e. over the marked simplex category).

Theorem. point 1. is a localisation of the (∞, n) -categories at the higher-exchanges, which does nothing when $n \leq 3$.

Conjecture. point 1. and 2. present the same model of higher-categories.

Features of the diagrammatic models

- Very strong **pasting theorems**;
- **Semi-strictification**: marked directed complexes may be used to two **equivalent** model:
 1. one where composition is witnessed by **the existence** of weak composites;
 2. one where composition is **an operation** satisfying strict associativity and exchange;

In both cases, the unit are algebraic and weak.

- Native support of **higher categorical constructions**, no need to use different presentation.
- A rich algebra of **weak units**.
- An overall feeling of a **topologically sound alternative** to strict n -categories

References

- C. Chanavat and A. Hadzihanovic, *Equivalences in diagrammatic sets*, Journal of Pure and Applied Algebra, 2026.
- A. Author, *Another reference*, Journal, year.